

0 2 . 1 There are different types of gene mutation.

Put a tick (✓) in the box next to the statement which describes **incorrectly** the effect of the mutation in an exon of a gene.

[1 mark]

A substitution may not result in a change to the encoded amino acid.

☐

An inversion will result in a change in the number of DNA bases.

☐

A deletion will result in a frame shift.

☐

An addition will result in a frame shift.

☐

0 2 . 2 Describe how alterations to tumour suppressor genes can lead to the development of tumours.

[3 marks]

0 2 . 3

A type of malignant tumour cell divides every 8 hours.

Starting with one of these cells, how many tumour cells will be present after 4 weeks?
Assume none of these cells will die.

Give your answer in standard form.

[2 marks]

Answer = _____

Turn over for the next question

